TOPIC

Assemble and Disassemble a PC Practical

Full hands-on session

Assemble and disassemble a desktop PC

Task: Document the process with labeled steps.

TASKS:1

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description for each.

RESULT

Main Components Inside a Desktop PC Cabinet

1. Motherboard

The motherboard is the main circuit board of the computer. It connects and allows communication between components such as the CPU, RAM, storage, and expansion cards.

2. CPU (Central Processing Unit)

The CPU is the main processor of the computer. It performs calculations and executes instructions required to run programs and applications.

3. RAM (Random Access Memory)

RAM is the temporary memory of the computer. It stores data and programs that are currently being used by the CPU.

4. CPU Cooler

The CPU cooler removes heat produced by the processor. It usually consists of a heatsink and a fan to keep the CPU at a safe temperature.

5. Power Supply Unit (PSU)

The PSU supplies electrical power to all the components inside the computer. It converts AC power from the wall socket into the required DC power.

6. Storage Drive (HDD/SSD)

The HDD or SSD stores the operating system, software, and personal files permanently. SSDs are generally faster than traditional HDDs.

7. Graphics Card (GPU)

The graphics card processes images, videos, and graphics for display on the monitor. A dedicated GPU is especially useful for gaming and other graphics-intensive tasks.

8. Case Fans

Case fans help circulate air inside the cabinet. They bring cool air in and remove warm air to prevent overheating.

9. PCIe Expansion Slots/Cards

PCIe slots are located on the motherboard and are used to install expansion cards such as graphics cards, sound cards, or network cards.

10. SATA Cables

SATA cables connect compatible storage drives such as HDDs and SATA SSDs to the motherboard. They transfer data between the storage drive and motherboard.

11. CMOS Battery

The CMOS battery is a small round battery on the motherboard. It helps maintain the BIOS/UEFI settings and system clock when the computer is turned off.

Conclusion

A desktop PC contains several important components that work together to operate the computer. The motherboard connects these components, while the CPU, RAM, storage, GPU, PSU, and cooling system perform different functions.

TASKS:2

Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC,

And

organize them in a document with captions explaining what you did at each step

Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard

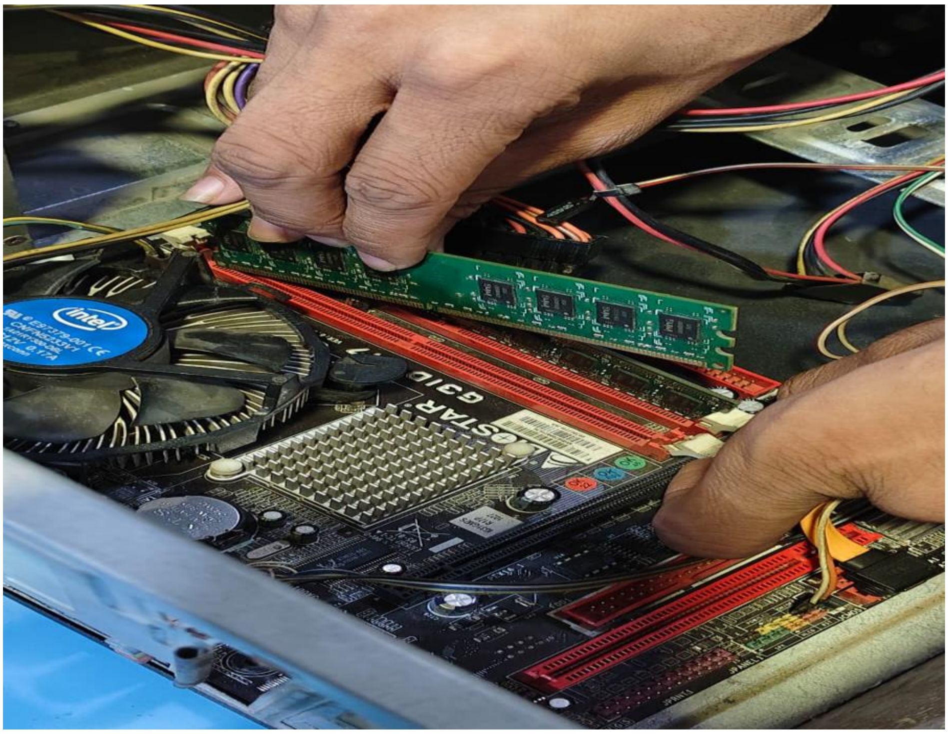
RESULT

Step 1: Shut Down and Unplug the PC

Step 2: Remove the Side Panel

Step 3: Disconnect Internal Cables

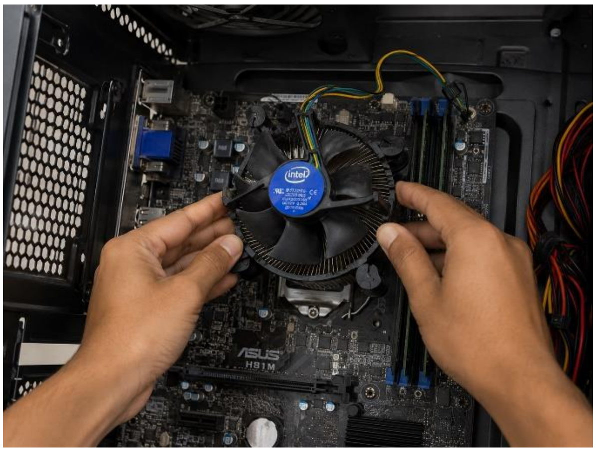
Step 4: Remove the RAM



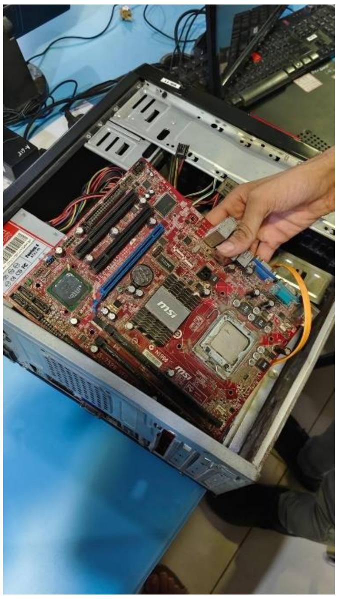
Step 5: Remove the HDD/SSD

Step 6: Remove the Graphics Card

Step 7: Remove the CPU Cooler



Step 8: Remove the Motherboard



Step 9: Final Disassembled Components

Safety Precautions

1. Always switch off and unplug the PC before opening the cabinet.
2. Handle electronic components carefully and hold them by their edges.
3. Do not force any connector or component.
4. Keep screws safely in one place.
5. Avoid touching the gold contacts and circuit-board components unnecessarily.

Conclusion

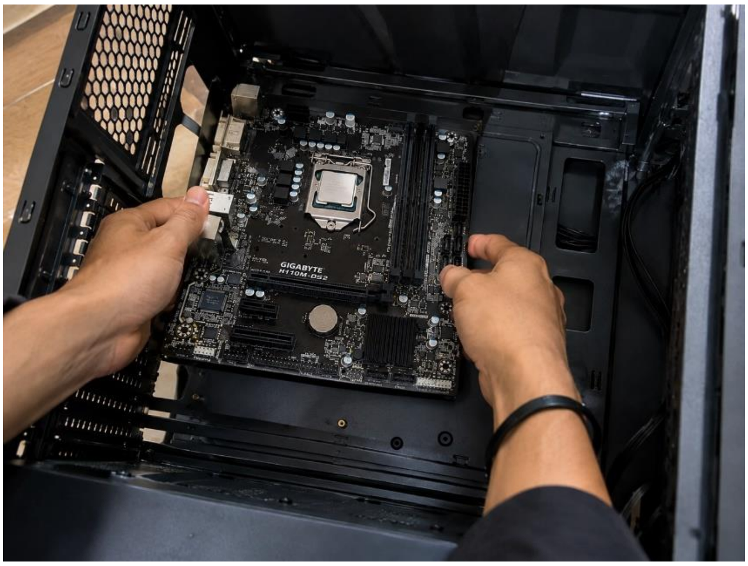
I successfully documented the step-by-step disassembly of a desktop PC. This activity helped me understand the location, connections, and functions of the major internal computer components.

TASKS:3

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

RESULT

Step 1: Install the Motherboard



What I did:
I carefully placed the motherboard inside the cabinet and aligned it with the screw holes and I/O opening. I then secured it with the appropriate screws.
Why this order matters:
The motherboard is the main platform for the other components, so it should be installed before connecting most internal components.

Step 2: Install the CPU Cooler

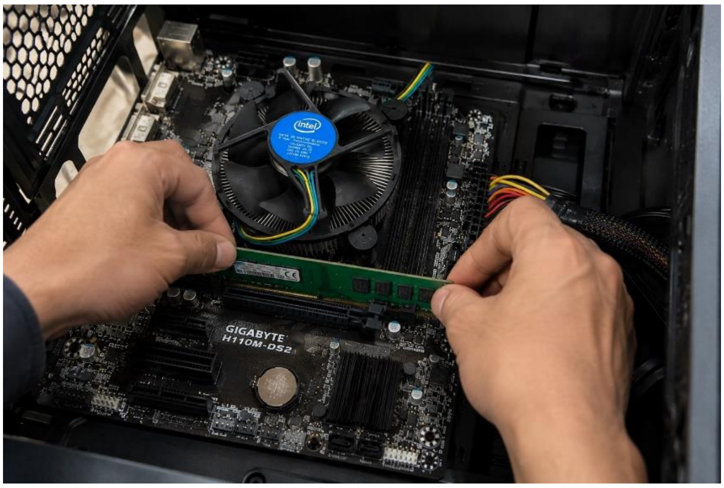
What I did:

I positioned the CPU cooler correctly over the CPU, secured its mounting mechanism, and connected the CPU fan cable to the CPU_FAN header.

Why this order matters:

The cooler must be properly attached before powering on the computer so the CPU can be cooled safely.

Step 3: Install the RAM



What I did:
I aligned the notch on the RAM module with the slot and pressed it down evenly until the retaining clips locked into place.
Why this order matters:
RAM must be properly seated so the computer can start and access its temporary memory.

Step 4: Install the Graphics Card

What I did:
I aligned the graphics card with the PCIe slot and pressed it down gently until it was properly seated. I then secured it to the case and connected its power cable if required.
Why this order matters:
The graphics card needs to be firmly seated in the correct PCIe slot for proper display and graphics performance.

Step 5: Install the HDD/SSD

What I did:

I placed the storage drive in its mounting position and secured it. I connected the SATA data cable to the motherboard and the power cable from the PSU.

Why this order matters:

The storage drive needs both a data connection and power connection to operate.

Step 6: Connect the Power Cables

What I did:
I connected the main motherboard power connector, CPU power connector, storage power cables, and graphics-card power cable if required.

Why this order matters:
All components need their required power connections before the PC can be tested.

Step 7: Check All Connections

What I did:
I checked that the RAM, graphics card, storage cables, power connectors, and fan cables were properly connected. I also checked that no loose screws or cables were interfering with the fans.

Why this order matters:
A final inspection helps prevent connection problems or damage before powering on the PC.

Step 8: Close the Cabinet

What I did:

I carefully placed the side panel back onto the cabinet and tightened the screws.

Why this order matters:

The side panel protects the internal components and helps maintain proper airflow inside the case.

Step 9: Reconnect External Cables and Power On

What I did:

I connected the monitor, keyboard, mouse, network cable if required, and power cable. I then switched on the PC and pressed the power button.

Why this order matters:

The PC should only be powered after all internal components and cables have been checked.

Conclusion

I successfully reassembled the desktop PC by reinstalling the motherboard, CPU, CPU cooler, RAM, graphics card, storage drive, and cables. Following the correct order helped ensure that each component was installed safely and connected properly.

TASKS:4

Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

RESULT

A. Safety Precautions

✓	SAFETY PRECAUTION	WHT IT IMPORTANT
☐	Shut down the PC completely	Prevents electrical damage and protects the components.
☐	Unplug the power cable	Disconnects the PC from mains electricity before opening the cabinet.
☐	Turn off the PSU switch	Provides an additional safety step before working inside the PC.
☐	Work on a clean, dry, flat surface	Prevents components from falling, getting wet, or being damaged.

✓	SAFETY PRECAUTION	WHT IT IMPORTANT
☐	Remove static electricity	Static discharge can damage sensitive components such as RAM and the motherboard.
☐	Handle components by their edges	Helps avoid touching electrical contacts and delicate circuits.
☐	Do not force connectors or components	Prevents broken pins, slots, and connectors.
☐	Keep screws organized	Makes reassembly easier and prevents missing screws.
☐	Take photos before disconnecting cables	Helps remember where each cable should be reconnected.
☐	Check all connections before powering on	Helps prevent loose connections and possible hardware problems.

B. Tools Needed

✓ Tool	Why It Is Important
☐ Phillips-head screwdriver	Used to remove and install most PC case and component screws.
☐ Anti-static wrist strap	Helps protect components from electrostatic discharge (ESD).
☐ Small container for SCREWS	Keeps screws safely organized during the process.
☐ Soft, clean cloth	Useful for cleaning dust from suitable external surfaces.
☐ Compressed air/dust blower	Helps remove dust from fans and other areas without touching components.

☐ Cable ties/Velcro straps	Helps organize cables after reassembly and improves airflow.
☐ Good lighting	Makes small connectors, screws, and motherboard labels easier to see.

Final Check Before Starting

☐ PC is completely switched off

☐ Power cable is disconnected

☐ Workspace is clean and dry

☐ Required tools are ready

☐ Components will be handled carefully

☐ Photos have been taken before disconnecting cable

Conclusion:

Following these precautions and using the correct tools helps protect both the person and the PC components. It also makes the assembly/disassembly process safer, more organized, and easier to complete.